ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	JGJB Gamage	**Registration No.**	D/BIT/24/0056
Degree / Intake	BIT / 41	**Group No.**	40
Project Title	AI-Powered Dynamic Public Transport Fleet Optimizer (Route 100)		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Frontend dashboard	Developed the depot planning board using Next.js App Router, TypeScript and Tailwind CSS, with a custom teal/ink theme and Fraunces / Manrope typography.	Next.js, React, TypeScript, Tailwind CSS	Full-page screenshot of the running dashboard (localhost:3000) at peak morning commute (8 min headway, bottleneck stops highlighted); Slides 11 & 13, Route100_Fleet_Optimizer_Group40.pptx
Planning controls	Built the input controls: direction, date, hour slider, weather, special event, and segmented Auto / override controls for day type, school vacation and school rush.	React hooks	Screenshot of scenario controls (Rain, Poya Day, Event Rush) with instant headway updates
Weather integration	Integrated Open-Meteo to prefill Clear / Rain for the selected date in Colombo, using forecast, archive and current-weather endpoints.	Open-Meteo API, fetch	Screenshot of the live Open-Meteo weather badge on the dashboard
Results visualisation	Implemented the KPI readout (headway, buses/hr, peak load, corridor ETA), crowd-status banner, corridor stop rail and demand-by-stop bar chart with peak-stop highlight.	Recharts, Tailwind CSS	Screenshot of the Corridor Rail with load progress bars and bottleneck badges; KPI cards (headway, ETA, bottleneck stop, load)
Backend API support	Connected the dashboard to POST /dispatch (payload, loading and error handling, request cancellation) and displayed calendar context chips and override hints; supported API setup and CORS configuration.	fetch, AbortController, FastAPI CORS	Code snippet from frontend/src/app/page.tsx showing the KPI state cards and the /dispatch API fetch handler

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.